```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv('/content/mall - mall.csv1.csv')
df.head(2)
```

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81

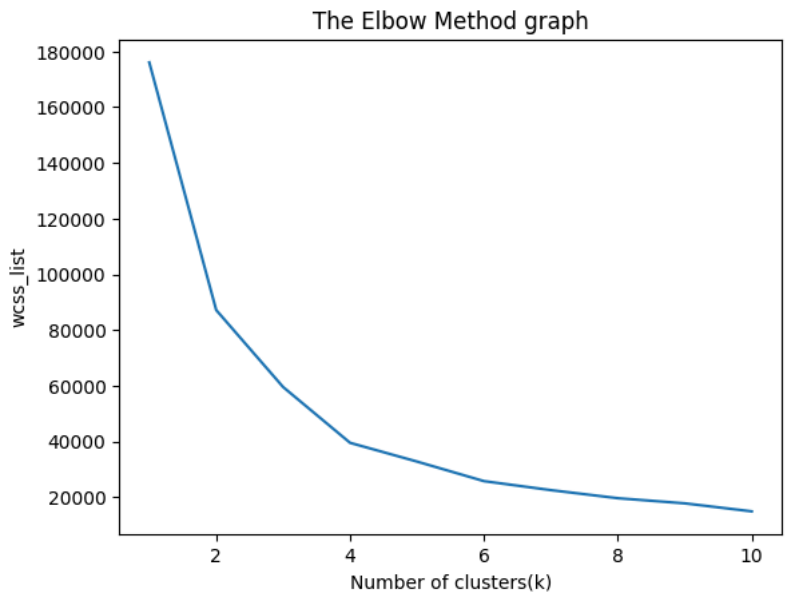
```
df=df.drop(columns=['CustomerID','Genre'])
```

```
x=df.iloc[:,[0,1]].values
```

```
from sklearn.cluster import KMeans
```

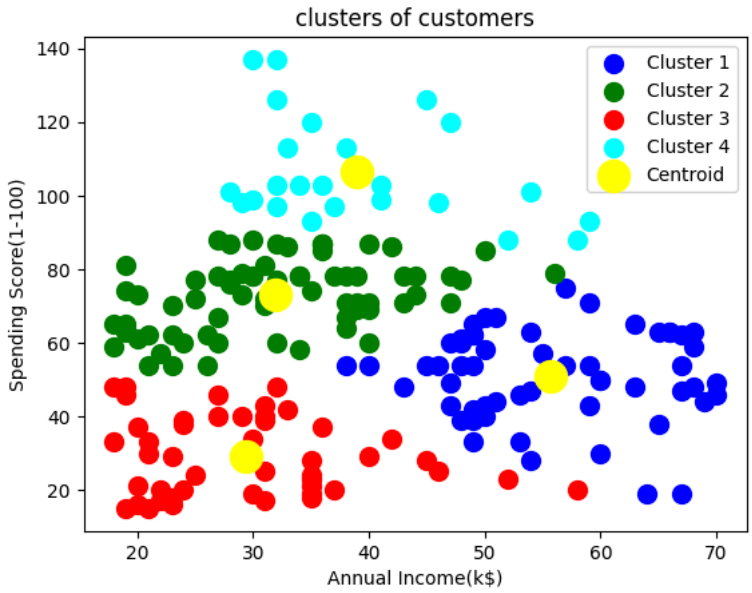
```
import matplotlib.pyplot as plt
```

```
a = []
for i in range(1,11):
  b=KMeans(n_clusters=i,init='k-means++',random_state=42)
  b.fit(x)
  a.append(b.inertia_)

plt.plot(range(1,11),a)
plt.title('The Elbow Method graph')
plt.xlabel('Number of clusters(k)')
plt.ylabel('wcss_list')                                # wcss = (Within Cluster Sum of Squares)
plt.show()
```




```
b = KMeans(n_clusters=4 , init='k-means++', random_state=42)
y_kmeans = b.fit_predict(x)
```

```
# visulaizing the clusters
plt.scatter(x[y_kmeans == 0, 0]  , x[y_kmeans == 0 ,1], s = 100, c= 'blue',label = 'Cluster 1')
plt.scatter(x[y_kmeans == 1, 0]  , x[y_kmeans == 1 ,1], s = 100, c= 'green',label = 'Cluster 2')
plt.scatter(x[y_kmeans == 2, 0]  , x[y_kmeans == 2,1], s = 100, c= 'red',label = 'Cluster 3')
plt.scatter(x[y_kmeans == 3, 0]  , x[y_kmeans == 3 ,1], s = 100, c= 'cyan',label = 'Cluster 4')

plt.scatter(b.cluster_centers_[:,0],b.cluster_centers_[:,1],s=300,c='yellow',label='Centroid')
plt.title('clusters of customers')
plt.xlabel('Annual Income(k$)')
plt.ylabel('Spending Score(1-100)')
plt.legend()
plt.show()
```



app user segmentation is a task of grouping users based on how they engage with the APP. IT HELPS find ratained users, find the user segment for a marketing campaign, and solve many others business problem where you need to search for user based on similar characteristics

```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv('/content/userbehaviour.....csv')
df.head(2)
```

	userid	Average Screen Time	Average Spent on App (INR)	Left Review	Ratings	New Password Request	Last Visited Minutes	Status
0	1001	17	634	1	9	7	2990	Installed
1	1002	0	54	0	4	8	24008	Uninstalled

```
import plotly.graph_objects as go
import plotly.express as px
import plotly.io as pio
import pandas as pd
pio.templates.default = "plotly_white"
```

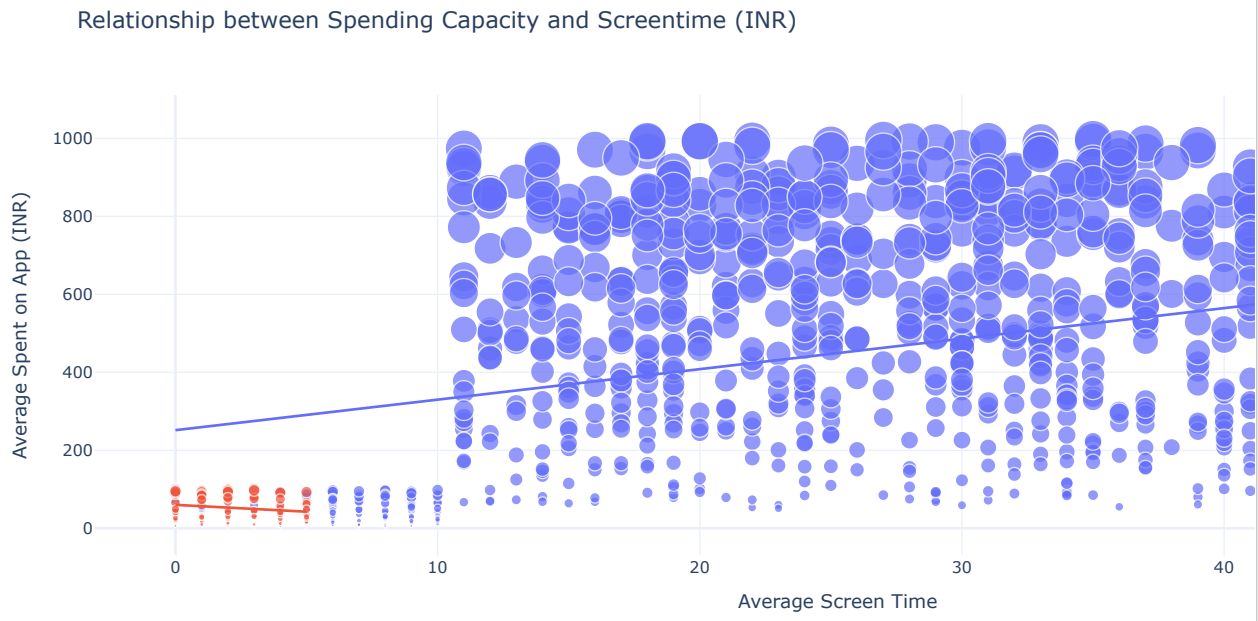
```
print(f"average screen time = {df['Average Screen Time'].mean()}")
print(f"highest screen time ={df["Average Screen Time"].max()}")
print(f"lowest screen time ={df["Average Screen Time"].min()}")
```

```
average screen time = 24.39039039039039
highest screen time =50
lowest screen time =0
```

```
print(f"averag spend of the users ={df["Average Spent on App (INR)"].mean()}")
print(f"highest spend of the users ={df["Average Spent on App (INR)"].max()}")
print(f"lowest spend of the users ={df["Average Spent on App (INR)"].min()}")
```

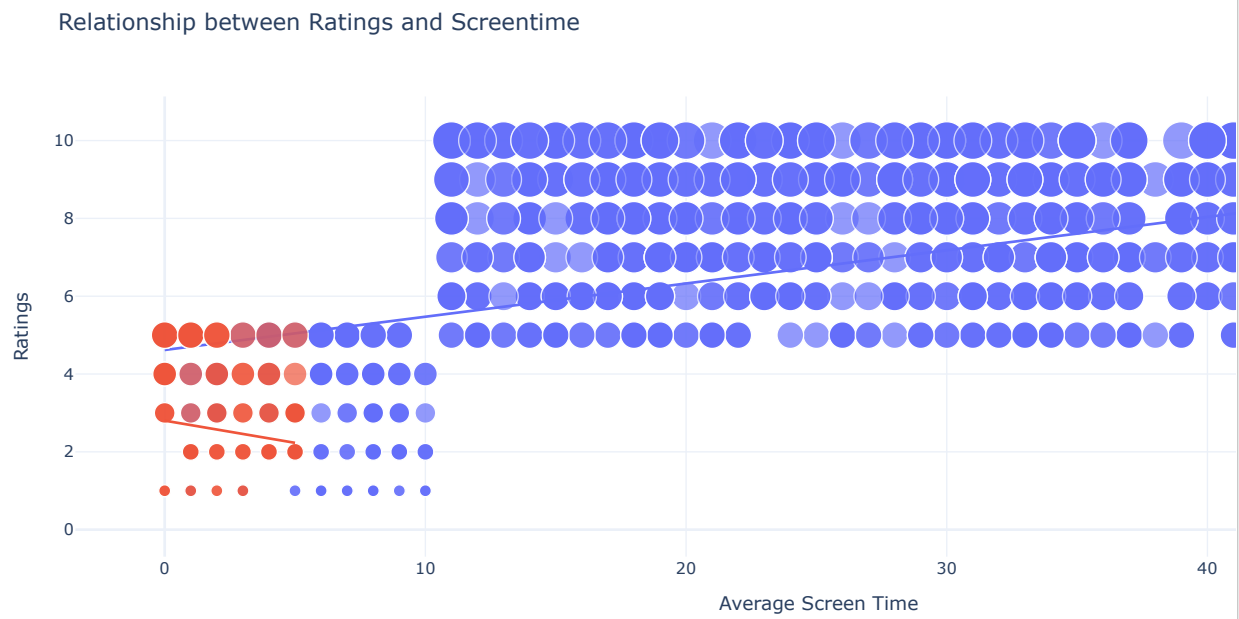
```
averag spend of the users =424.4154154154154
highest spend of the users =998
lowest spend of the users =0
```

```
figure=px.scatter(data_frame=df,x="Average Screen Time",
                  y="Average Spent on App (INR)",
                  size="Average Spent on App (INR)",
                  color="Status",
                  title="Relationship between Spending Capacity and Screentime (INR)",
                  trendline="ols")
figure.show()
```



Double-click (or enter) to edit

```
figure=px.scatter(data_frame=df,x="Average Screen Time",
                  y="Ratings",
                  size="Ratings",
                  color="Status",
                  title="Relationship between Ratings and Screentime",
                  trendline="ols")
figure.show()
```




```
clustering_data=df[["Average Screen Time","Left Review",
                    "Ratings","Last Visited Minutes",
                    "Average Spent on App (INR)",
                    "New Password Request"]]
```

```
clustering_data
```

	Average Screen Time	Left Review	Ratings	Last Visited Minutes	Average Spent on App (INR)	New Password Request
0	17	1	9	2990	634	7
1	0	0	4	24008	54	8
2	37	0	8	971	207	5
3	32	1	6	799	445	2
4	45	1	5	3668	427	6
...	...	...	...	...	...	...
994	38	0	5	1865	938	4
995	43	0	6	1327	61	8
996	47	0	6	1936	761	1
997	6	1	5	3436	73	1
998	9	0	4	4039	54	2

999 rows × 6 columns

```
from sklearn.preprocessing import MinMaxScaler
for i in clustering_data.columns:
  MinMaxScaler(i)

from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(clustering_data)
df['segments'] = clusters

df.head(2)
```

```
df['segments'] = df['segments'].map({0: "retained", 1: "churn", 2: "need attention"})
```

```
df.head(2)
```

	userid	Average Screen Time	Average Spent on App (INR)	Left Review	Ratings	New Password Request	Last Visited Minutes	Status	segments
0	1001	17	634	1	9	7	2990	Installed	retained
1	1002	0	54	0	4	8	24008	Uninstalled	churn

Start coding or generate with AI.

1000
800
NR)
0
1
2